Linear Search

1. Source Code (C)

```c
#include <stdio.h>

// Function for linear search
int linearSearch(int arr[], int n, int key) {
    for (int i = 0; i < n; ++i) {
        if (arr[i] == key) {
            return i;
            // Return the index if element is found
        }
    }

    // Return -1 if element is not found
    return -1;
}

int main() {
    int arr[] = {12, 45, 67, 23, 56, 89, 9, 43};
    int n = sizeof(arr) / sizeof(arr[0]);
    int key, index;

    printf("\n--Linear Search--\n");
    printf("Enter the element to search: ");
    scanf("%d", &key);

    index = linearSearch(arr, n, key);

    if (index != -1) {
        printf("Element %d found at position %d.\n", key, index + 1);
    }
    else {
        printf("Element %d not found in the array.\n", key);
    }

    return 0;
}
```

2. Program Output

```
--Linear Search--

Enter the element to search: 45
Element 45 found at position 2.

--Linear Search--

Enter the element to search: 11
Element 11 not found in the array.
```